

D1	Understand and use the relationship between roots and coefficients of polynomial equations up to quartic equations.
	Content
D2	Form a polynomial equation whose roots are a linear transformation of the roots of a given polynomial equation (of at least cubic degree).
	Content
D3	Understand and use formulae for the sums of integers, squares and cubes and use these to sum other series.
	Content
D4	Understand and use the method of differences for summation of series including use of partial fractions.
	Content
D5	Find the Maclaurin series of a function including the general term.
	Content
D6	Recognise and use the Maclaurin series for e^x , $\ln(1+x)$, $\sin x$, $\cos x$, and $(1+x)^n$, and be aware of the range of values of x for which they are valid (proof not required).
	Content
D7	Evaluation of limits using Maclaurin series or l'Hôpital's rule.
	Content
D8	Inequalities involving polynomial equations (cubic and quartic).
	Content
D9	Solving inequalities such as $\frac{ax+b}{cx+d} < ex + f$ algebraically.
	Content
D10	Modulus of functions and associated inequalities.
	Content
D11	Graphs of $y = f(x) $, $y = \frac{1}{f(x)}$ for given $y = f(x)$

	Content
D13	Graphs of rational functions of form $\frac{ax^2 + bx + c}{dx^2 + ex + f}$, including cases when some of these coefficients are zero; asymptotes parallel to coordinate axes; oblique asymptotes.

	Content
D14	Using quadratic theory (not calculus) to find the possible values of the function and coordinates of the stationary points of the graph for rational functions of form $\frac{ax^2 + bx + c}{dx^2 + ex + f}$

	Content
D15	Sketching graphs of curves with equations $y^2 = 4ax$, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $xy = c^2$ including intercepts with axes and equations of asymptotes of hyperbolas.

	Content
D16	Single transformations of curves involving translations, stretches parallel to coordinate axes and reflections in the coordinate axes and the lines $y = \pm x$. Extend to composite transformations including rotations and enlargements.